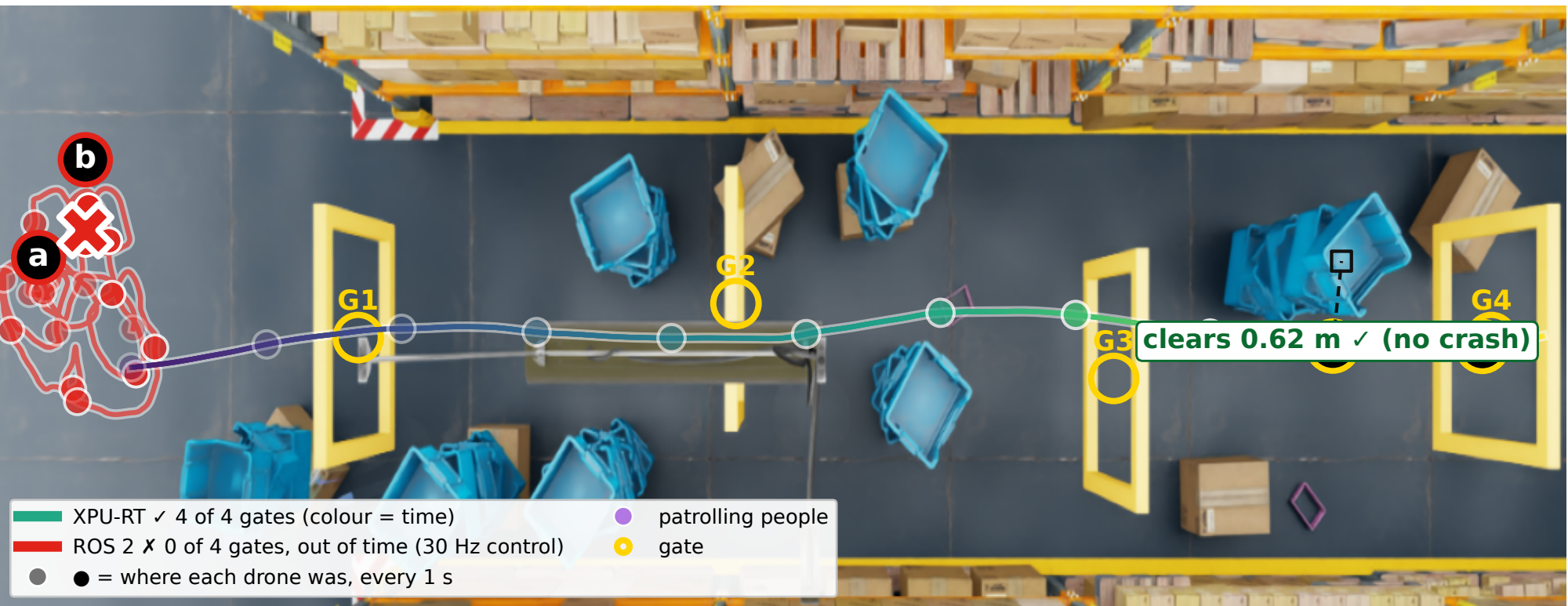


Warehouse gate-course showdown — same scene and drone, each arm at gain 0.5 / its own control rate, at 1.4 m/s

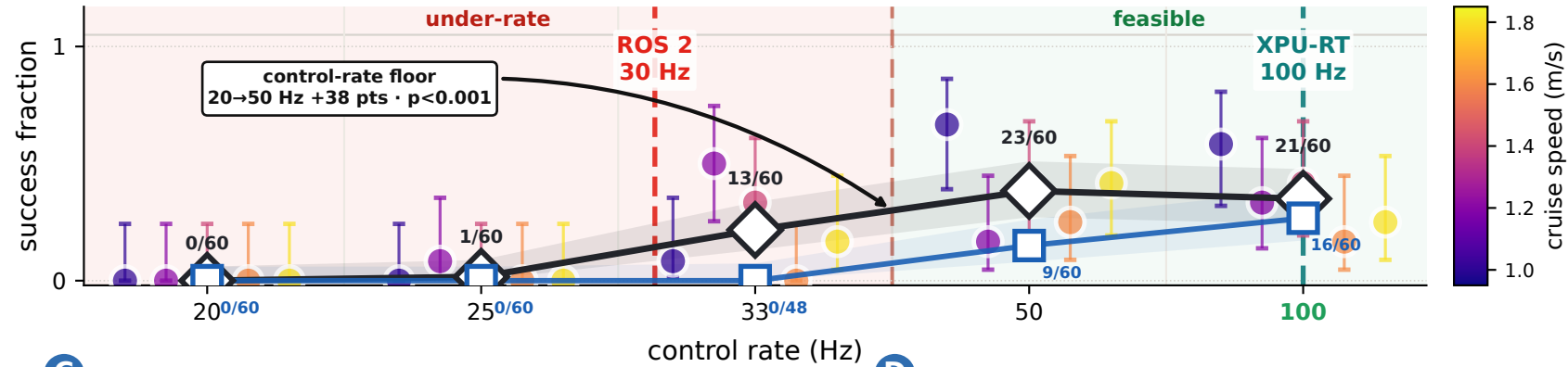
XPU-RT · solver-placed CP-SAT schedule (100 Hz control) clears all 4 gates

ROS 2 · 4-hart YOLO pool, 4-wide nav, all 8 cores (30 Hz control) runs out of time short of G1

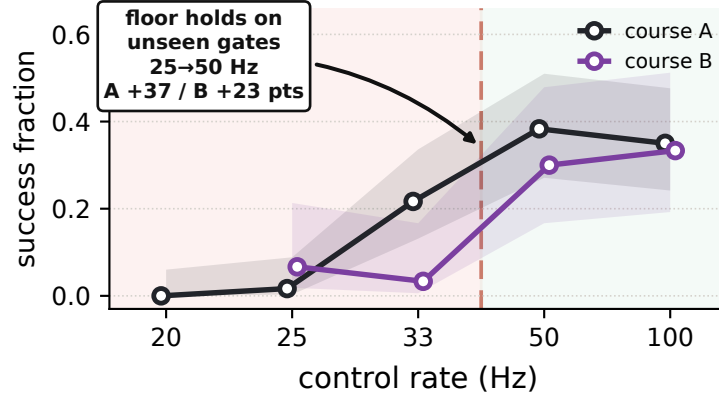
one seed of this scene (seed 1007); over its 12 seeds neither arm completes — 1.8 vs 0.1 of 4 gates



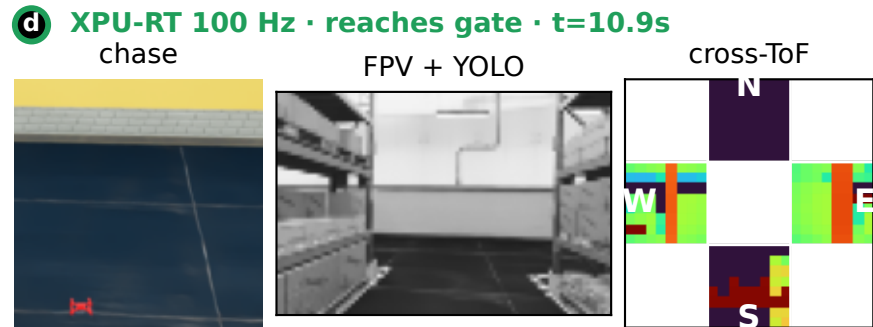
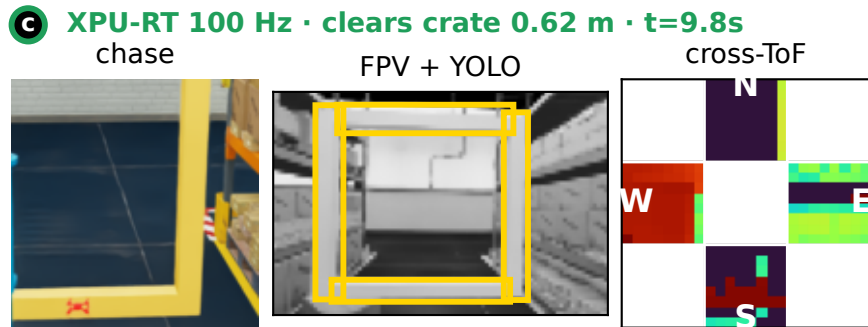
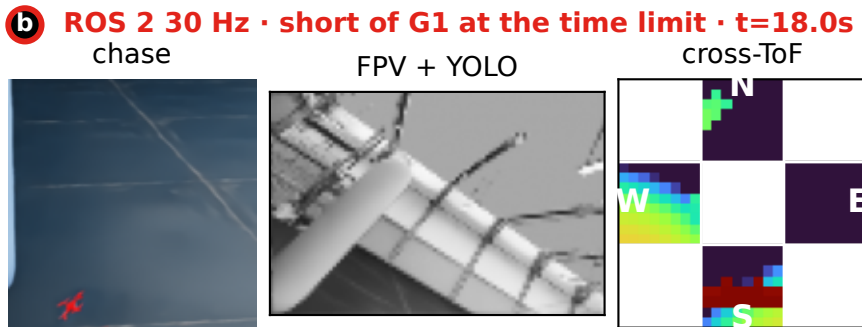
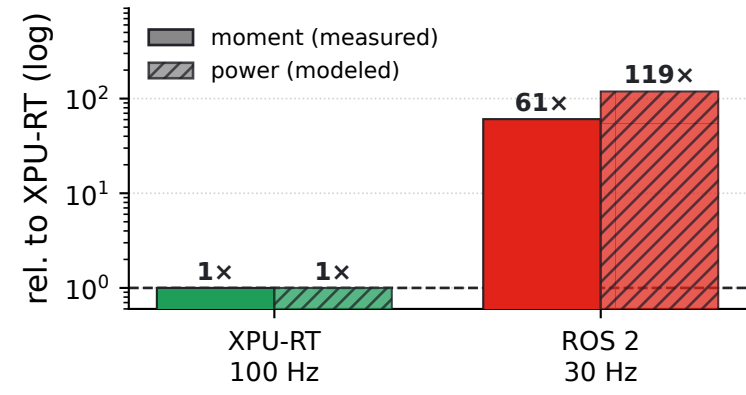
B Control-rate envelope (rate injected, 300 flights): success against control rate — ROS 2's 30 Hz and XPU-RT's 100 Hz marked



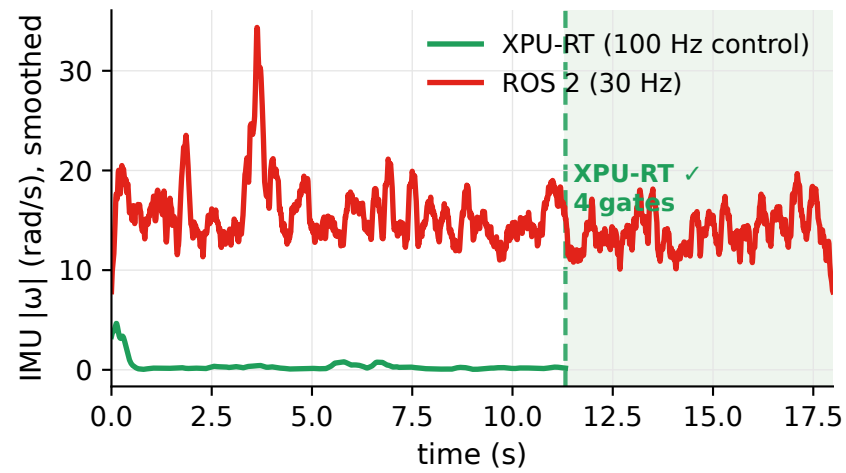
C Generalization: unseen gates (course B)



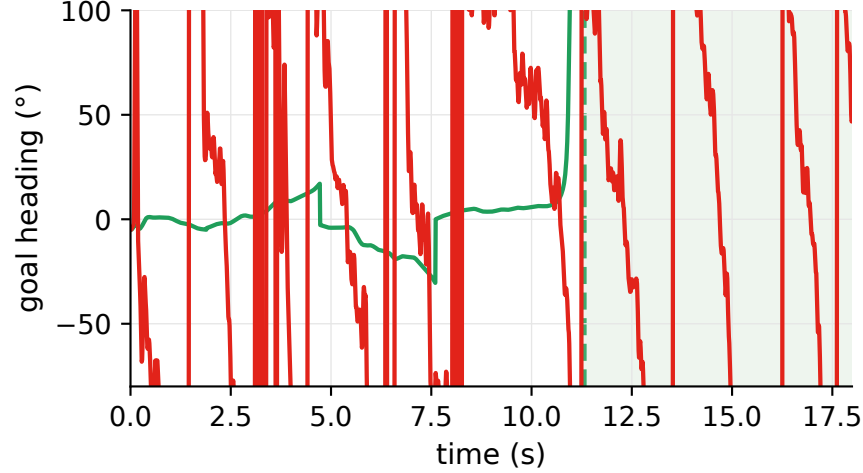
D Mechanism: the baseline thrashes (n=6 flights/arm)



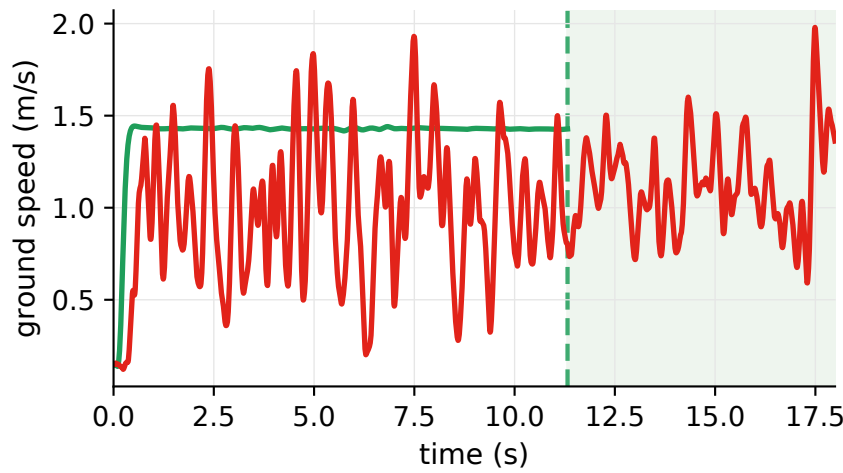
E Inside the pair — body rate: the baseline thrashes at 30 Hz



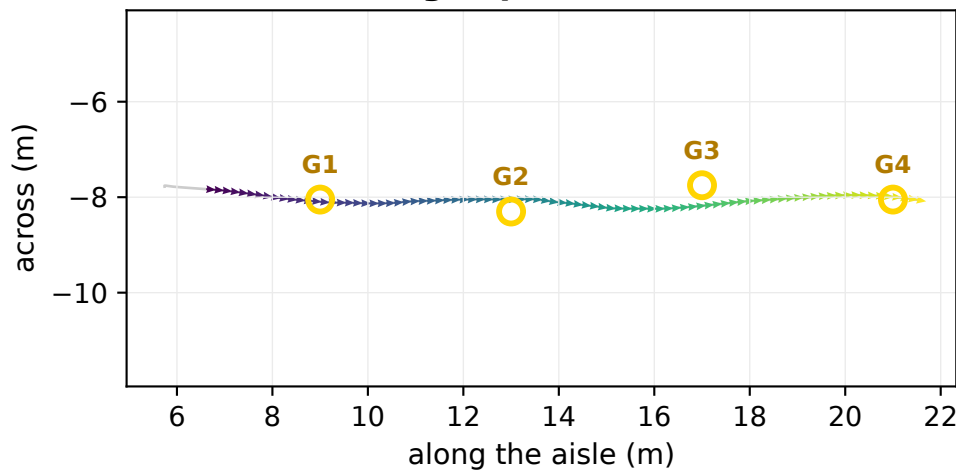
F Nav goal heading: XPU-RT's settles, the baseline's never does



G Forward speed: XPU-RT holds the cruise, the baseline never reaches it



H XPU-RT's velocity through the gates (arrow = heading · speed, colour = time)



I Onboard K1 schedule, measured at the 30 Hz camera — XPU-RT · solver-placed CP-SAT schedule places the kernels over all 8 harts: camera→control 27 ms, control every 10.0 ms, every frame on time · YOLO on 8 harts, 5 traced, nav on 8 harts, 3 traced, control on 1 hart
ROS 2 · all 8 cores (3 processes · single executor · chained control · pinned): camera→control 30 ms, control every 33.4 ms, 0/598 frames late · YOLO on 4 harts, nav on 4 harts, 1 traced, control on 1 hart

